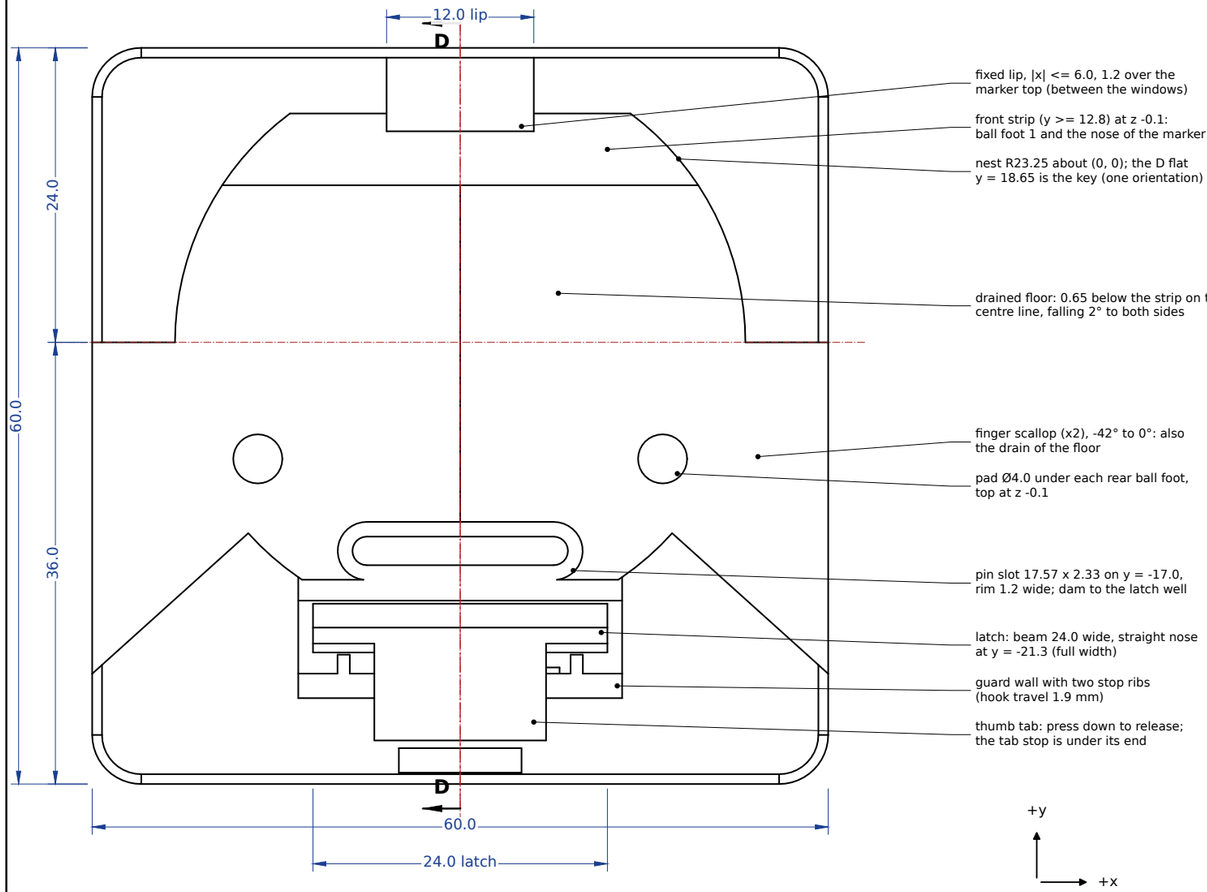
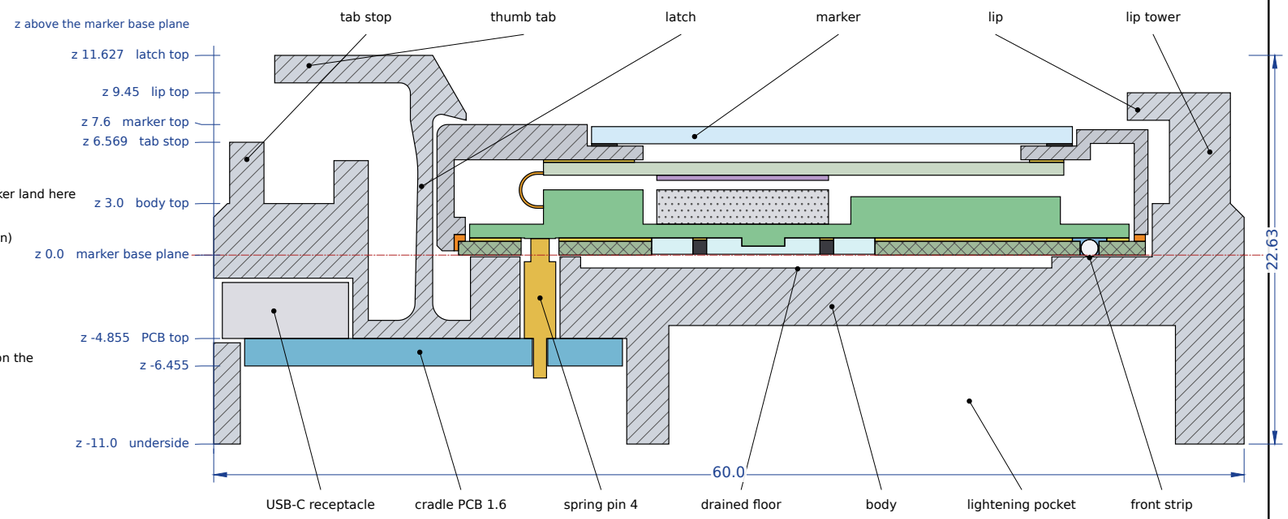
TOP VIEW 2:1



SECTION D-D (x = 0), marker seated, centre pin on its pad 2.8:1



VIEW FROM BELOW 1.5:1 (+x to the LEFT)

NOTES

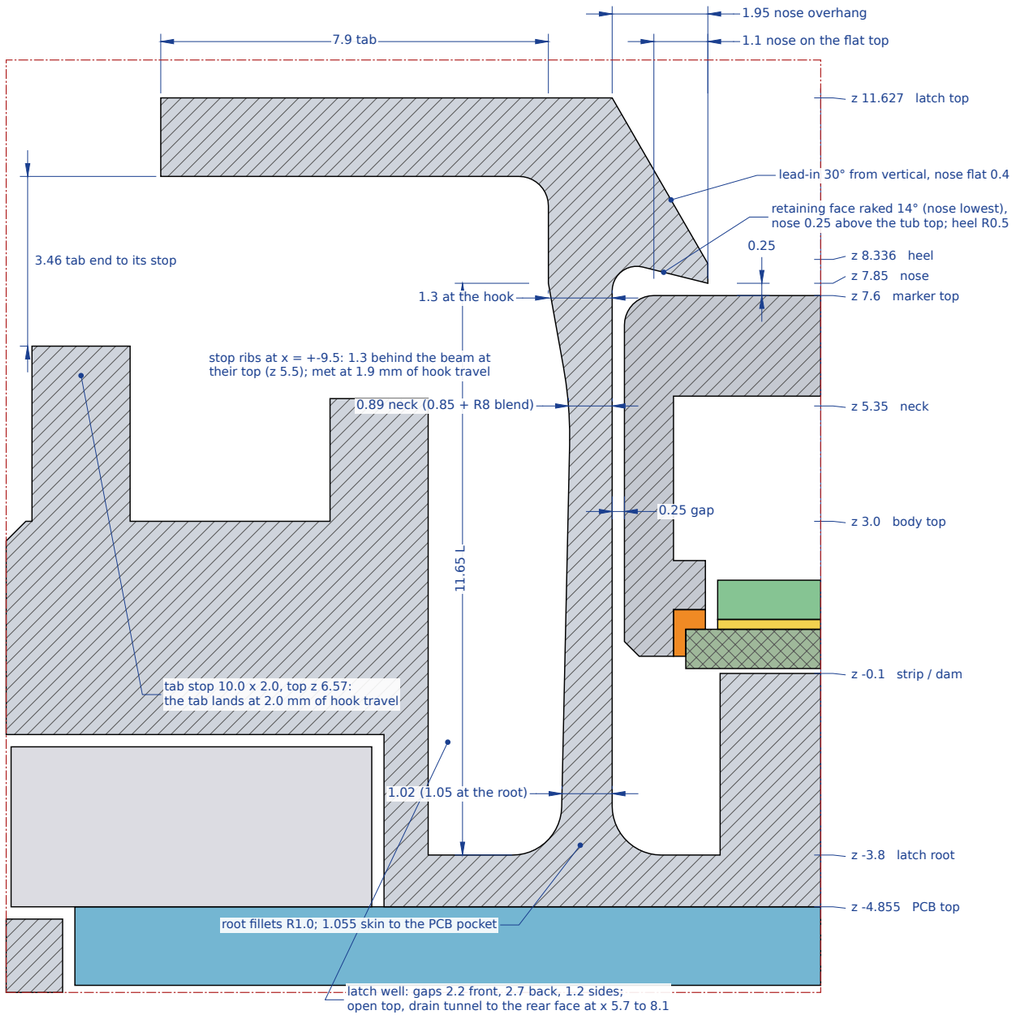
1. Charging and data dock only (rev D: every number of the YAML cradle block). Printed body 60.0 x 60.0 x 14.0 mm (z -11.0 to 3.0); lip and latch stand above it, overall height 22.63 mm. Frame: the marker frame with the marker seated (origin at the marker centre on its base plane); body from y -36.0 to 24.0.
2. The nest is the marker outline + 0.25 mm; the D flat is the key, so the marker seats one way round only. Its three ball feet rest on the front strip and on two pads (z = -0.1); the rest of the floor is a basin that drains through the scallops. A rim round the pin slot and a dam in front of the latch well keep water off the cradle PCB; the well drains through a tunnel (2.4 x 2.4) to the rear face beside the USB opening.
3. Hold-down: fixed lip over the top front edge (|x| <= 6.0, clear of windows, cover glass and vent) + sprung latch at the rear with a thumb tab. The marker goes in nose first under the lip and is pressed down at the rear until the latch clicks. Details and numbers: sheet 2.
4. Cradle PCB top at z = -4.855: the spring pins stand at their working height (5.82) on the copper face of the pads (z 0.965). Seven pin positions, consumer build fits four (VBUS, D-, D+, GND at x = -7.62 ... 0.0), factory fixture seven. Pin and receptacle envelopes are PROVISIONAL.
5. PRINT ORIENTATION LOCKED: the x axis of this drawing is the build direction (cradle_print.stl / .step), so that the profiles of latch and lip lie in the build plane. No tumbling, no vapor smoothing. No wall below 0.8 mm, latch root 1.05 mm, no trapped powder (checks 5, 6).
6. First-article dimensions: front nest wall to hook nose 39.95 +-0.15; beam to stop ribs 1.3; tab-stop top z 6.57.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

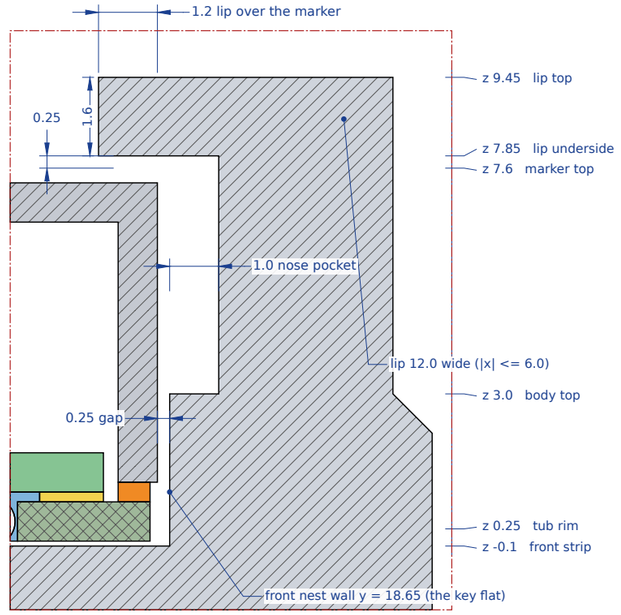
TITLE
GL1-020 Cradle (charging / data dock with hold-down)

PROJECT / PART		
GreenLine GL-1 electronic ball marker - cradle body		
MATERIAL / FINISH	PROCESS	
PA12 (HP 3D HR PA 12), any colour, not smoothed, not tumbled	HP Multi Jet Fusion, orientation LOCKED (note 5); drill 4 x Ø1.6	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	2:1 / 2.8:1 / 1.5:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	1 of 2 2026-11-13

DETAIL J - rear latch, section x = 0 8:1

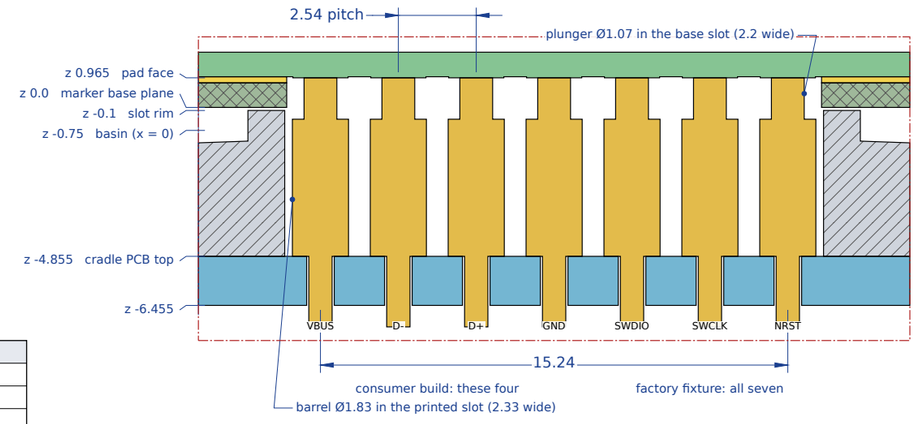


DETAIL K - front lip, section x = 0 8:1

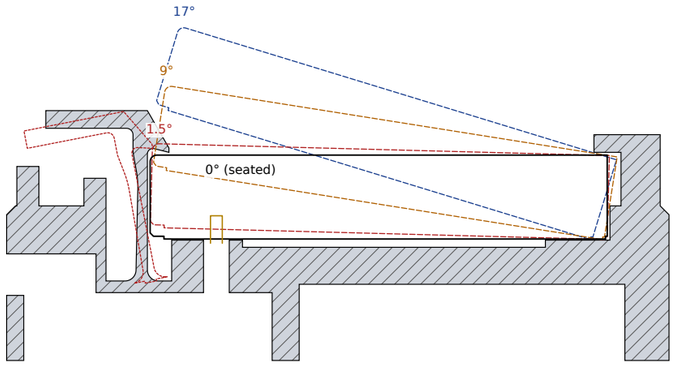


Cradle PCB (2 layers)	Value
Outline	30.0 x 22.0 x 1.6, x +-15.0, y -34.2 to -12.2
Pin positions (7)	y = -17.0, x = -7.62 ... 7.62, pitch 2.54
Screws (4 x M2 thread-forming)	(+-12.5, -31.5) and (+-12.5, -14.5)
USB-C receptacle	GCT USB4105, top face, mouth at y -35.5
Top face	z = -4.855 (pad face 0.965 - working height 5.82)

SECTION E-E (y = -17.0) pin row, marker seated 5:1



INSERTION PATH 1.8:1 section x = 0, marker outline at four tilts



Hold-down (PA12 MJF, E = 1700 MPa); check 7	Value
Latch beam: b; h root / neck / at the hook; L	24.0; 1.05 / 0.85 (0.89 blended) / 1.3; 11.65 mm
Pin force: 7 pins / 4 pins; marker weight	4.13 N / 2.36 N; 0.12 N
Retention, no friction, marker centred	31.65 N at the hook = 8.8 x the 7-pin force
marker against the front wall (governing)	21.65 N = 6.0 x (>= 3 x); needs E >= 848 MPa
governing, overlap -0.3 mm (tolerance)	12.36 N = 3.4 x; needs E >= 1485 MPa
Hook travel: insertion / release fwd, ctr, rear	1.42 / 1.37, 1.59, 1.86 mm (posed solids)
Full deflection: stop ribs / tab stop	1.90 / 2.00 mm of hook travel
Strain there: beam formula (FE peak); limit	1.81 (2.05) / 1.78 (1.99) %; 2.5 %
Insertion force at the thumb (mu 0.2 / 0.3 / 0.4)	9.5 / 11.4 / 13.7 N
Release: thumb on the tab (fwd / ctr / rear)	8.8 / 10.2 / 11.9 N; tab lands at 12.9 N
Marker float against lip and hook with pins fitted	up to 0.25 mm (pins then 6.07 mm high)

NOTES

1. Tilt the marker nose-down, slide the top front edge under the lip, lower the rear: its bottom edge runs down the 30° lead-in, pushes the latch back by up to 1.42 mm (red, dotted) and the hook snaps over the top.
2. The pads meet the free plungers (tip z 2.11, orange) at about 1.9° of tilt; seated, the pins are compressed 1.14 of 2.29 mm.
3. Release: press the thumb tab down until the hook lets go (the tab stop limits it); the pins lift the rear about 1.3 mm; lift at the finger scallops.
4. Check 7 poses the marker solids at 19 tilts from 17° to 0°: no interference with lip, nest walls, floor or pins.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

GL1-020 Cradle (charging / data dock with hold-down)

PROJECT / PART

GreenLine GL-1 electronic ball marker - hold-down details, pin row, insertion

MATERIAL / FINISH	PROCESS
PA12 (HP 3D HR PA 12), any colour, not smoothed, not tumbled	see sheet 1
GENERAL TOLERANCE	SCALE (A3) / UNITS
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	8:1 / 5:1 / 1.8:1 mm

STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	2 of 2 2026-11-13